

- Neonates exposed to SSRIs or SNRIs late in the third trimester have developed complications requiring prolonged hospitalization, respiratory support, and tube feeding; reported symptoms are consistent with either a direct toxic effect of SSRIs and SNRIs or, possibly, a drug discontinuation syndrome, and include respiratory distress, cyanosis, apnea, seizures, temperature instability, feeding difficulty, vomiting, hypoglycemia, hypotonia, hypertonia, hyperreflexia, tremor, jitteriness, irritability, and constant crying
- National Pregnancy Registry for Antidepressants: 1-866-961-2388, <https://womensmentalhealth.org/research/pregnancyregistry/antidepressants>

Breast Feeding

- Some drug is found in mother's breast milk
- Trace amounts may be present in nursing children whose mothers are on paroxetine
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued
- Immediate postpartum period is a high-risk time for depression, especially in women who have had prior depressive episodes, so drug may need to be reinstituted late in the third trimester or shortly after childbirth to prevent a recurrence during the postpartum period
- Must weigh benefits of breast feeding with risks and benefits of antidepressant treatment versus nontreatment to both the infant and the mother

- For many patients, this may mean continuing treatment during breast feeding

The Art of Psychopharmacology

Potential Advantages

- Patients with anxiety disorders and insomnia
- Patients with mixed anxiety/depression

Potential Disadvantages

- Patients with hypersomnia
- Alzheimer/cognitive disorders
- Patients with psychomotor retardation, fatigue, and low energy

Primary Target Symptoms

- Depressed mood
- Anxiety
- Sleep disturbance, especially insomnia
- Obsessions, compulsions
- Panic attacks, avoidant behavior, re-experiencing, hyperarousal

* Often a preferred treatment of anxious depression as well as major depressive disorder comorbid with anxiety disorders

* Withdrawal effects may be more likely than for some other SSRIs when discontinued (especially akathisia, restlessness, gastrointestinal symptoms, dizziness, tingling, dysesthesias, nausea, stomach cramps, restlessness)

- Inhibits own metabolism, so dosing is not linear

* Paroxetine has mild anticholinergic actions that can enhance the rapid onset of anxiolytic and hypnotic efficacy but also cause mild anticholinergic side effects

- Can cause cognitive and affective “flattening”
- May be less activating than other SSRIs
- Paroxetine is a potent CYP450 2D6 inhibitor
- SSRIs may be less effective in women over 50, especially if they are not taking estrogen
- SSRIs may be useful for hot flashes in perimenopausal women
- Some anecdotal reports suggest greater weight gain and sexual dysfunction than some other SSRIs, but the clinical significance of this is unknown
- For sexual dysfunction, can augment with bupropion, sildenafil, tadalafil, or switch to a non-SSRI such as bupropion or mirtazapine
- Some postmenopausal women’s depression will respond better to paroxetine plus estrogen augmentation than to paroxetine alone

- Nonresponse to paroxetine in elderly may require consideration of mild cognitive impairment or Alzheimer disease
- CR formulation may enhance tolerability, especially for nausea
- Can be better tolerated than some SSRIs for patients with anxiety and insomnia and can reduce these symptoms early in dosing

Suggested Reading

Bourin M , Chue P , Guillon Y . Paroxetine: a review . *CNS Drug Rev* 2001 ;**7** :25 –47 .

Gibiino S , Serretti A. Paroxetine for the treatment of depression: a critical update . *Expert Opin Pharmacother* 2012 ;**13** (3):421–31 .

Green B . Focus on paroxetine . *Curr Med Res Opin* 2003 ;**19** :13 –21 .

Wagstaff AJ , Cheer SM , Matheson AJ , Ormrod D , Goa KL . Paroxetine: an update of its use in psychiatric disorders in adults . *Drugs* 2002 ;**62** :655 –703 .

Perospirone

Therapeutics

Brands

- Lullan
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine and serotonin receptor antagonist (DS-RAn)
- Atypical antipsychotic (serotonin-dopamine antagonist, second-generation antipsychotic)

Commonly Prescribed for

(bold for FDA approved)

- Schizophrenia (Japan)

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis

- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
- * Interactions at 5HT1A receptors may contribute to efficacy for cognitive and affective symptoms in some patients

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition and affective stabilization
- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients require up to 16–20 weeks to show a good response, especially on cognitive symptoms

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenic patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year

- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Some patients may require treatment with a conventional antipsychotic
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection

- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse

Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation of perospirone has not been systematically studied
- Addition of a benzodiazepine, especially short-term for agitation
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may theoretically be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful

Tests

* Potential of weight gain, diabetes, and dyslipidemia associated with perospirone has not been systematically studied, but patients should be monitored the same as for other atypical antipsychotics

Before starting an atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile

- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)
 - has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese,

pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic

* Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, and clouding of sensorium, even coma

- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and perospirone should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia

- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent
- Receptor binding portfolio of perospirone is not well characterized

Notable Side Effects

- * Drug-induced parkinsonism, akathisia
- * Insomnia
- Sedation, anxiety, weakness, headache, anorexia, constipation
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- Elevated creatine phosphokinase levels



Life-Threatening or Dangerous Side Effects

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability

with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure

- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain

* Not well characterized

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia

- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Reduce the dose
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 8–48 mg/day in 3 divided doses

Dosage Forms

- Tablet 4 mg, 8 mg

How to Dose

- Begin at 4 mg 3 times a day, increasing as tolerated up to 16 mg 3 times a day



Dosing Tips

- Some patients have been treated with up to 96 mg/day in 3 divided doses
- Unknown whether dosing frequency can be reduced to once or twice daily, but by analogy with other agents in this class with half-lives shorter than 24 hours, this may be possible
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Not reported

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Slow down-titration (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after perospirone is discontinued

Pharmacokinetics

- Metabolized primarily by CYP450 3A4
- No active metabolites



Drug Interactions

- Ketaconazole and possibly other CYP450 3A4 inhibitors such as nefazodone, fluvoxamine, and fluoxetine may increase plasma levels of perospirone

- Carbamazepine and possibly other inducers of CYP450 3A4 may decrease plasma levels of perospirone

Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and perospirone should be used cautiously in patients at risk for aspiration pneumonia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If there is a proven allergy to perospirone

Special Populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Use with caution

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Some patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to

placebo, and also have an increased risk of cerebrovascular events



Children and Adolescents

- Use with caution



Pregnancy

- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary

Breast Feeding

- Unknown if perospirone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- In Japan, studies suggest efficacy for negative symptoms of schizophrenia

Potential Disadvantages

- Patients who have difficulty complying with three times daily administration

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Affective symptoms (depression, anxiety)
- Cognitive symptoms

Pearls

- Drug-induced parkinsonism may be more frequent than with some other atypical antipsychotics
- Potent 5HT1A binding properties may be helpful for improving cognitive symptoms of schizophrenia in long-term treatment
- Theoretically, should be effective in acute bipolar mania

Suggested Reading

Ohno Y . Pharmacological characteristics of perospirone hydrochloride, a novel antipsychotic agent . *Nippon Yakurigaku Zasshi* 2000 ;**116** (4):225–31 .

Perphenazine

Therapeutics

Brands

- Trilafon
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: dopamine receptor antagonist (D-RAn)
- Conventional antipsychotic (neuroleptic, phenothiazine, dopamine 2 antagonist, antiemetic)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia**
- **Nausea, vomiting**
- Other psychotic disorders
- Bipolar disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis
- Combination of dopamine D2, histamine H1, and cholinergic M1 blockade in the vomiting center may reduce nausea and vomiting

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but may take several weeks for full effect on behavior
- Injection: initial effect after 10 minutes, peak after 1–2 hours
- Actions on nausea and vomiting are immediate

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Continue treatment in schizophrenia until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis in schizophrenia
- For second and subsequent episodes of psychosis in schizophrenia, treatment may need to be indefinite

- Reduces symptoms of acute psychotic mania but not proven as a mood stabilizer or as an effective maintenance treatment in bipolar disorder
- After reducing acute psychotic symptoms in mania, switch to a mood stabilizer and/or an atypical antipsychotic for mood stabilization and maintenance

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics
- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation of conventional antipsychotics has not been systematically studied
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful
- Addition of a benzodiazepine, especially short-term for agitation

Tests

* Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

* Monitor weight and BMI during treatment

* Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics

* While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic

- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit
- Phenothiazines may cause false-positive phenylketonuria results

- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and perphenazine should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus